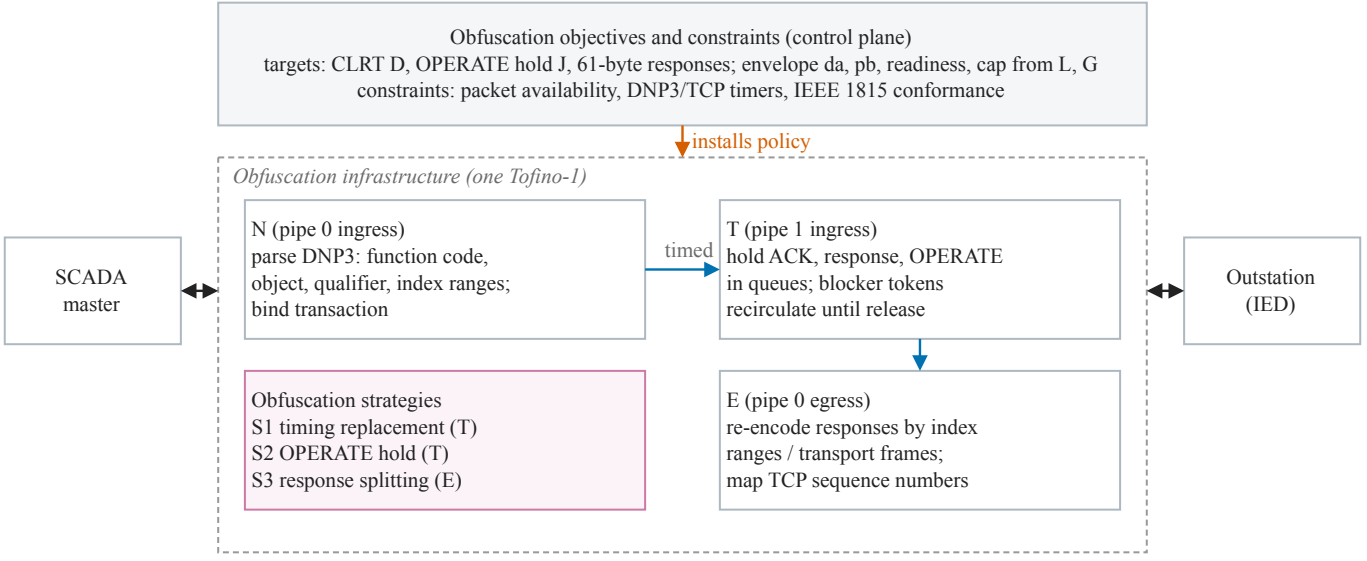
Obfuscation objectives and constraints (control plane)
targets: CLRT D, OPERATE hold J, 61-byte responses; envelope da, pb, readiness, cap from L, G
constraints: packet availability, DNP3/TCP timers, IEEE 1815 conformance
installs policy
Obfuscation infrastructure (one Tofino-1)
SCADA master
N (pipe 0 ingress)
parse DNP3: function code, object, qualifier, index ranges; bind transaction
timed
T (pipe 1 ingress)
hold ACK, response, OPERATE in queues; blocker tokens recirculate until release
Outstation (IED)
Obfuscation strategies
S1 timing replacement (T)
S2 OPERATE hold (T)
S3 response splitting (E)
E (pipe 0 egress)
re-encode responses by index ranges / transport frames; map TCP sequence numbers